LABS PORTFOLIO · LAB 3 · WEEK 9

Architectural Decision Record

Resilience Patterns with an LLM API Circuit Breaker

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Title: Fail-fast fallback for an unavailable LLM API

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Context

The `POST /generate` service depends on an external LLM API, which can have variable latency, become unavailable, or rate-limit requests. In a distributed system, an unresponsive dependency can cause a cascading failure: requests wait for timeouts, consume resources, and may cause upstream callers to time out too. Blind retries can add load to a struggling API and, for a paid LLM, waste tokens or money. The service therefore needs to bound each call, handle transient failures, and stop sending requests when failures indicate a continuing outage. It must also give callers a clear fallback rather than appear to generate an answer when it cannot reach the model.

Decision

Combine a 30-second timeout, bounded retry with exponential backoff, and a circuit breaker. Retry eligible failures, including HTTP 429 (rate limit) and 503 (service unavailable), for a maximum of three attempts, with an initial one-second delay and multiplier of two; the delay grows exponentially as retries continue. Do not retry authentication errors, which require correcting credentials or configuration. The timeout bounds how long an individual LLM call can occupy the service. In CLOSED state, requests are forwarded normally and outcomes are recorded in a rolling window of ten. After five failures in that window, transition to OPEN. In OPEN state, fail fast: do not call the LLM API, and return HTTP 503 with `{"status":"circuit_open","retry_after":10,"message":"LLM service temporarily unavailable"}`. After ten seconds, transition to HALF-OPEN and allow a trial request. If it succeeds, close the circuit and resume normal traffic; if it fails, reopen the circuit. A call that times out returns HTTP 504. Do not serve cached or fabricated model output as the fallback.

Consequences

A transient failure may add retries and backoff delay before a result or error is returned; the breaker records the final outcome of the request rather than each attempt as a separate request. The timeout prevents one slow call from waiting indefinitely. During a sustained outage, callers receive a prompt, explicit 503 rather than waiting on an API call known to be failing; they can show an unavailable message and use `retry_after` to guide a later retry. The distinction is useful: 503 means the circuit is OPEN and the API call was not attempted, while 504 means an attempted call exceeded its timeout. The service cannot generate new answers until the LLM recovers, so availability of `/generate` is reduced during the outage. The circuit breaker limits unnecessary requests to the failing dependency, reducing pressure and potential repeated API costs. The HALF-OPEN trial provides a controlled way to test recovery before normal traffic resumes. Retry logs, circuit-state transitions, and Prometheus metrics make these behaviors observable in the chaos tests. Retry delay and the fixed threshold are trade-offs; production settings should be tuned to real latency, traffic, and provider limits.

Alternatives

Returning an error immediately on the first failure would be simple and fast, but would expose transient errors that a retry could resolve. Unbounded retrying could eventually succeed, but risks long waits, resource exhaustion, cascading failures, and repeated charges; bounded exponential backoff limits that risk. A cached response could provide some continuity, but this service does not maintain a cache, and a response to an earlier prompt may be stale or inappropriate for the current request. A generic or partially generated response would misleadingly imply that the requested model operation succeeded. The selected 503 fallback is explicit and truthful: unlike stale or invented content, it tells the caller the dependency is unavailable, while the breaker and HALF-OPEN trial support recovery.